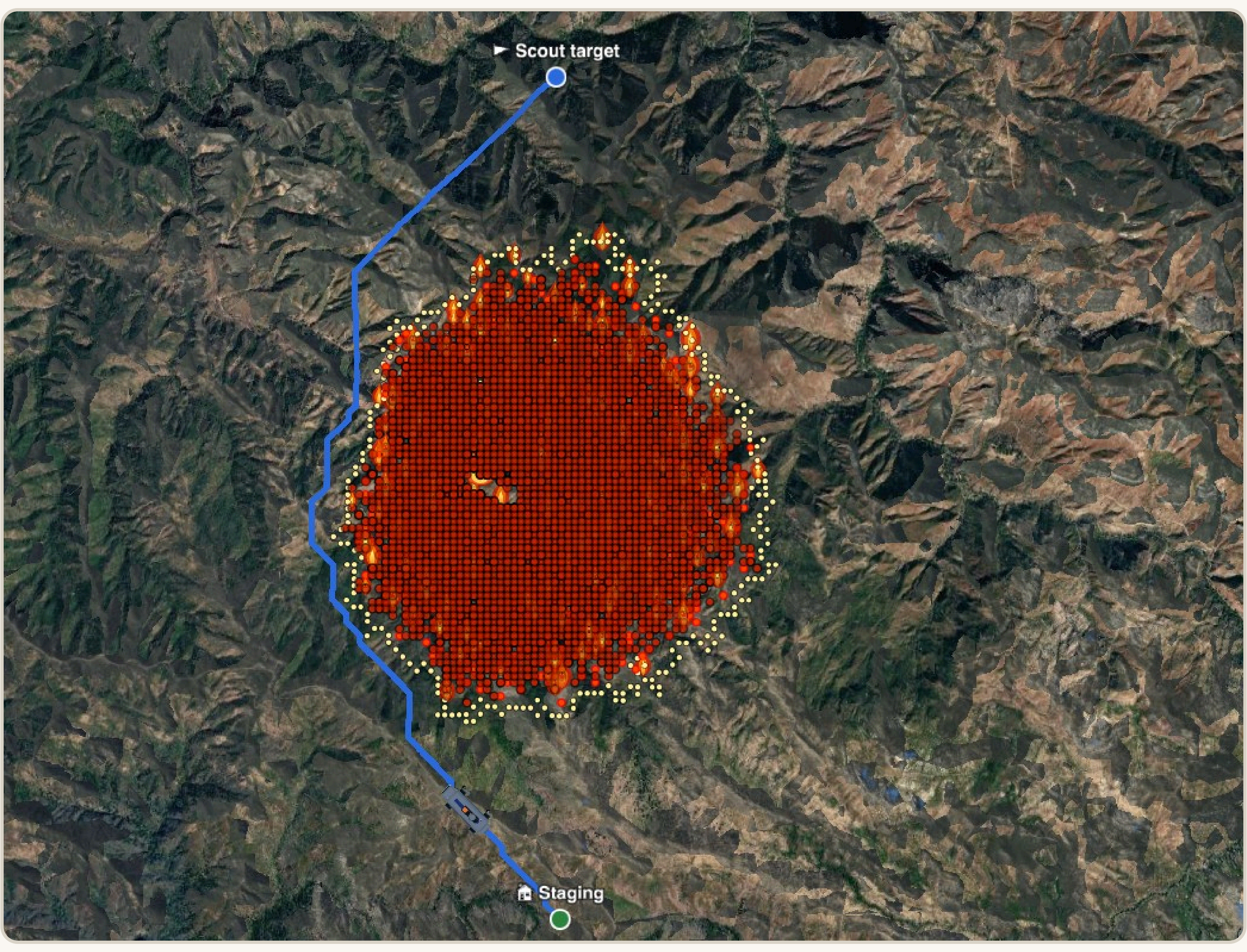
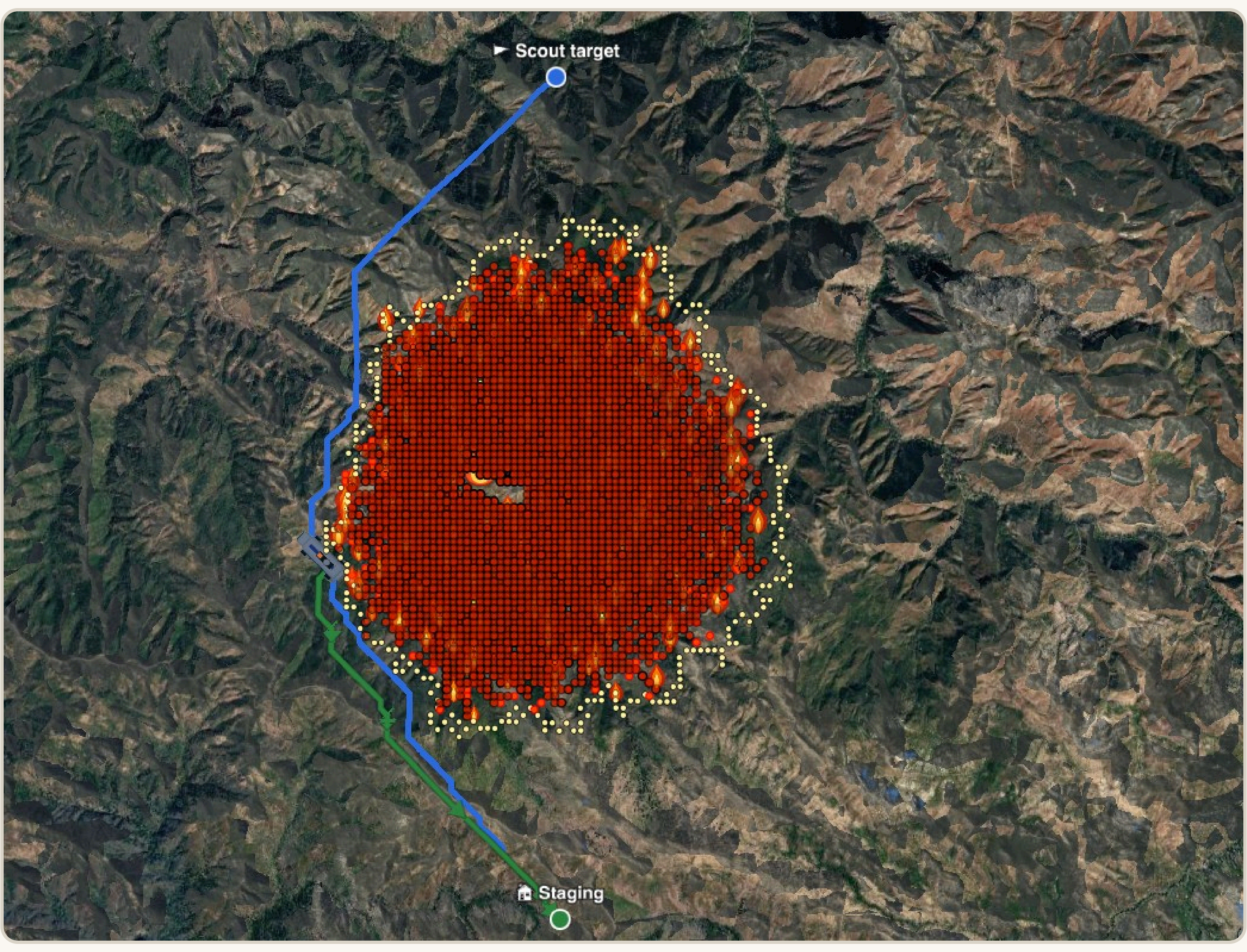
Robot dog in simulation

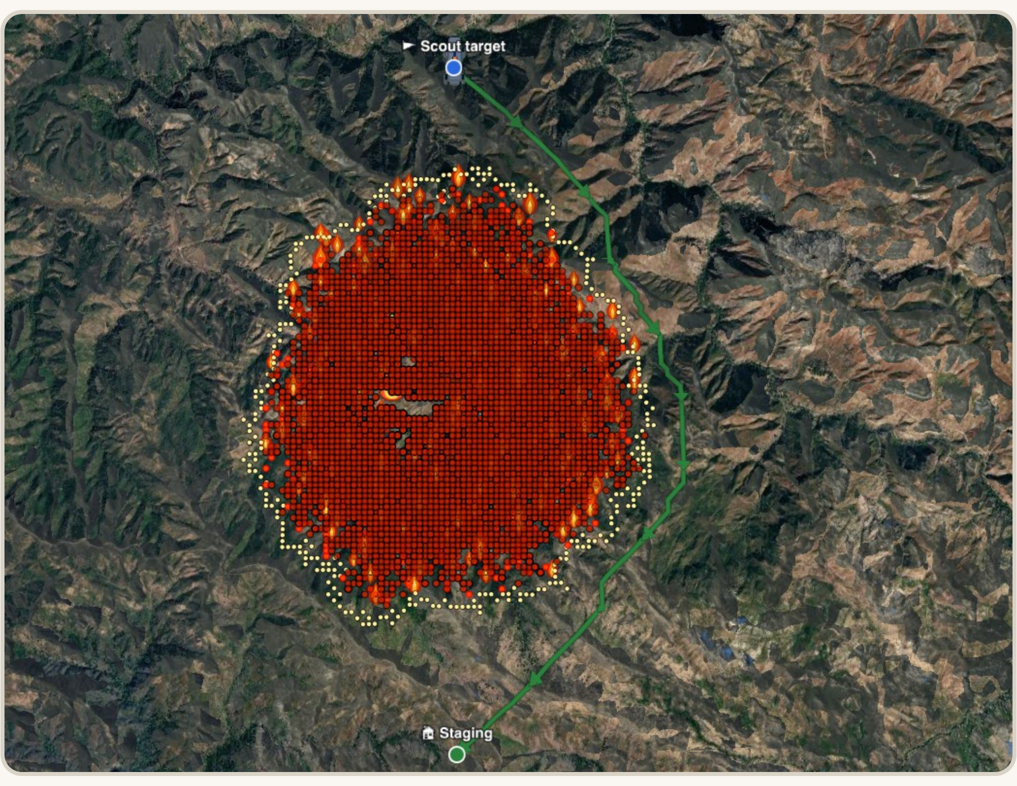
Mock simulation on the real forecast – no physical robot. A hidden "true" fire runs on the real local weather; the dog only knows the forecast until it reports.


Scout target
Staging

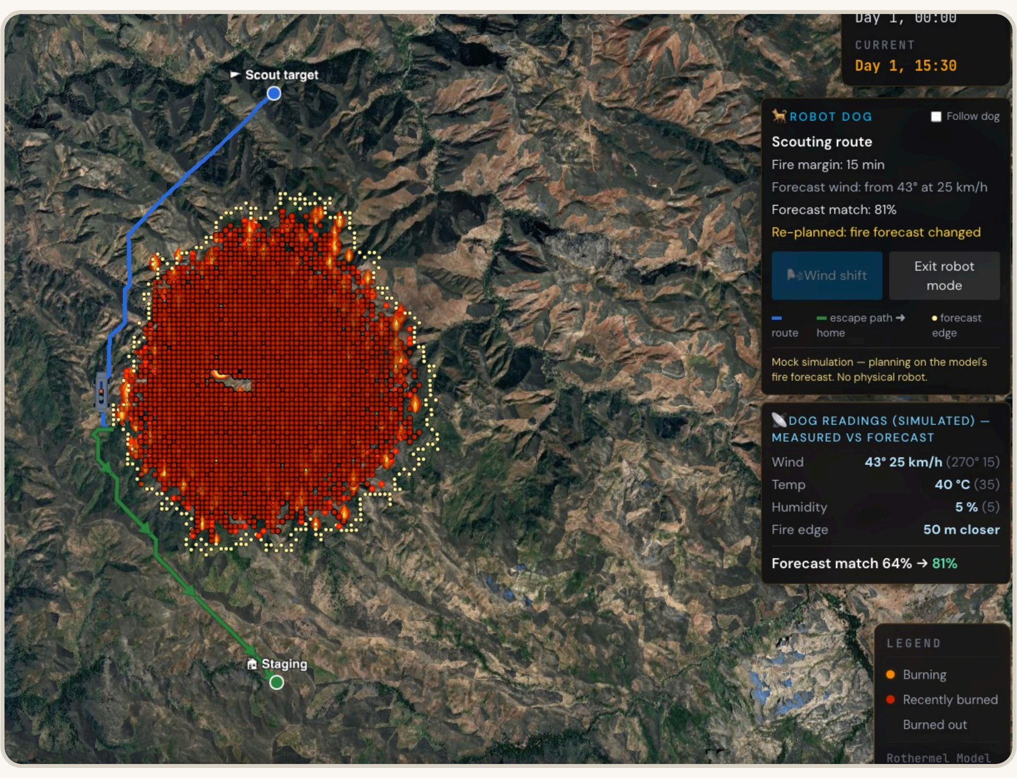
② **Safe route.** The route (**blue**) bends around cells the forecast says will ignite before the dog gets through. Yellow dots: the forecast's fire edge.


Scout target
Staging

③ **Escape path.** A wind shift turns the true fire toward the dog – the forecast doesn't know yet. The escape path (**green**, arrows pointing home) always stays open.


Scout target
Staging

④ **Scout.** The dog reaches the fire edge, scouts it, and walks home before the fire arrives.


Scout target
Staging
☐ Follow dog
Wind shift
Exit robot mode
route
escape path → home
● forecast edge
LEGEND
● Burning
● Recently burned
Burned out
Rothermel Model

Day 1, 00:00

CURRENT
Day 1, 15:30

ROBOT DOG

Scouting route
Fire margin: 15 min
Forecast wind: from 43° at 25 km/h
Forecast match: 81%
Re-planned: fire forecast changed

Mock simulation – planning on the model's fire forecast. No physical robot.

DOG READINGS (SIMULATED) – MEASURED VS FORECAST

Wind	**43° 25 km/h** (270° 15)
Temp	**40 °C** (35)
Humidity	**5 %** (5)
Fire edge	**50 m closer**

Forecast match 64% → 81%

⑤ **Feedback.** The dog sends measured wind, temperature, humidity and the real fire edge. Forecast match for the next 4 h of spread: **64% → 81%**; the dog re-plans.

Status & next

- ✔ ML fire forecast trained and tested
- ✔ 3D browser simulator on real terrain
- ✔ Containment planner (search)
- ✔ Robot-dog loop (route, escape, scout, feedback) in mock simulation
- → Run forecast + planner on the dog or its controller
- → Real sensors feeding the forecast
- → Scout the planned containment line
- → Tests on real robot dogs

Open-source code + demo video

Data & references

Code: github.com/sxz2012/wildfire-simulation
NASA FIRMS (VIIRS active-fire detections) · Open-Meteo weather archive · ESA WorldCover 10 m land cover · Cesium World Terrain · Google Photorealistic 3D Tiles
Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.
Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*